

C871

# MA, Science Education Teacher Performance Assessment

Armored Drill

8 concepts · 12 items · 4 transfer items

**What this is.** Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

**Source.** Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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## How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

|                             |                                           |                        |                                  |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>_____ | [ ] sure [ ] unsure [ ]<br>guess |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|

# Items

1.

A teacher planning a six-week ecology unit notes that the classroom has four working microscopes for 28 students, the school is in a suburb with a large park nearby, and the school mascot is a hawk. Which contextual-factors statement best meets the task?

- A.** The hawk mascot will be used as the unit's anchor organism because students identify with it.
- B.** With four microscopes for 28 students, labs will run as rotating stations in groups of seven, and the nearby park will serve as a field site for data collection.
- C.** The class has 28 students, four microscopes, and a park nearby.
- D.** Because equipment is limited, all labs will be replaced with teacher demonstrations.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

2.

A teacher with ten years of experience drafts goals for a unit on forces aligned to a middle school standard. The unit will use toy cars borrowed from a colleague. Which goal is best written?

- A.** Students will plan and carry out an investigation to show how the mass of a car affects its change in motion under the same push.
- B.** Students will understand Newton's second law.
- C.** Students will memorize  $F = ma$ .
- D.** Students will race toy cars down ramps.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

3.

For a unit on chemical reactions in a building renovated last summer, the pre-assessment asks students to balance equations and identify evidence of a chemical change. The draft post-assessment asks students to name lab safety equipment. The unit runs four weeks. What is the main problem?

- A.** The post-assessment should be shorter than the pre-assessment.
- B.** The pre-assessment should be dropped because students know little at the start.
- C.** There is no problem as long as both are scored out of 100.
- D.** The post-assessment does not measure the same goals as the pre-assessment, so gains cannot be attributed to the unit.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**4.** [TRANSFER — reversed stem / different variable]

A teacher's class scored much higher on the post-assessment than on the pre-assessment. Which fact would most weaken the claim that students learned the unit goals?

- A.** The post-assessment was given in the afternoon.
- B.** The same items were used on both tests.
- C.** Some students still scored below 70% on the post-test.
- D.** The post-assessment covered easier, recall-level items than the pre-assessment.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**5.**

A teacher, who shares a lab room with two colleagues, drafts a density lesson: day 1, notes and vocabulary; day 2, worked examples; day 3, a lab where students float and sink objects to confirm the formula. Which revision best fits inquiry-based design?

- A.** Skip the notes entirely; students never need formal vocabulary.
- B.** Open with the float-and-sink exploration, have students form explanations from their data, then introduce the formula and vocabulary.
- C.** Move the lab to day 2 so it fits the shared-room schedule.
- D.** Keep the order but add a quiz after the notes.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**6.** [TRANSFER — reversed stem / different variable]

Students have just explored how the length of a pendulum affects its swing time and recorded patterns. In a 5E lesson, what should come next?

- A.** An end-of-unit test (Evaluate).
- B.** Students share patterns and the teacher helps them build a formal explanation (Explain).
- C.** Applying the idea to a new system, such as a swing (Elaborate).
- D.** A hook to get students interested (Engage).

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

7.

During a unit on seasons, taught in a room with a new interactive whiteboard, many students write that summer happens because Earth is closer to the Sun. What is the most effective next step?

- A. Tell students the correct answer and move on to the next topic.
- B. Use the whiteboard to show a video about the Moon's phases.
- C. Show data that the Northern and Southern Hemispheres have opposite seasons at the same Earth-Sun distance, and have students revise their models.
- D. Accept the idea, since it is partly true.

|                             |                                           |                        |                                  |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>_____ | [ ] sure [ ] unsure [ ]<br>guess |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|

8.

Halfway through a unit on cells, a teacher who teaches three sections a day gives an exit ticket. 18 of 26 students confuse the cell wall with the cell membrane; the rest are secure. The pacing guide calls for starting genetics tomorrow. What is the best instructional decision?

- A. Reteach the wall-versus-membrane distinction with a targeted activity for the 18, give the 8 an extension, then recheck before moving on.
- B. Start genetics as planned to stay on pace.
- C. Reteach the entire cells unit to all 26 students.
- D. Give the same exit ticket again tomorrow.

|                             |                                           |                        |                                  |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>_____ | [ ] sure [ ] unsure [ ]<br>guess |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|

9. [TRANSFER — reversed stem / different variable]

A teacher's unit report says, "On day 6, I added a second modeling activity on energy transfer." What must accompany this statement to show instructional decision-making?

- A. The cost of the materials used
- B. A statement that the change always improves learning
- C. The student evidence that prompted the change and how students did afterward
- D. The original lesson plan only

|                             |                                           |                        |                                  |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>_____ | [ ] sure [ ] unsure [ ]<br>guess |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|

10.

A teacher who runs the school garden club analyzes a unit. Class pre-test mean: 35%; post-test mean: 78%. English learners (6 students): 30% to 60%. Other students (20): 37% to 83%. Which analysis is best supported?

- A. The class gained 78 points.
- B. The unit worked equally well for all students because the class mean rose.
- C. English learners learned the least because they scored lowest on the post-test.
- D. The class gained 43 points, but English learners gained 30 points versus 46 for peers, suggesting language supports need strengthening.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

11. [TRANSFER — reversed stem / different variable]

A class gained an average of 40 percentage points and scored 80% on the post-assessment. What was the pre-assessment average?

- A. 120%
- B. 50%
- C. 40%
- D. 80%

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

12.

After teaching a unit in a portable classroom with a noisy heater, a teacher drafts a reflection. The data show strong gains on the modeling goal but weak gains on the goal about designing controlled experiments. Which reflection best meets the task?

- A. Students gained least on experimental design; item analysis shows they rarely held variables constant. Next time I will add a guided fair-test activity, and I will study inquiry scaffolds with my department.
- B. The unit went well overall, so no changes are needed.
- C. The heater noise explains the weak results.
- D. Students did not try hard on the experiment goal.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

# STOP

Answers and dissection follow. Finish every item and every strip first.

# Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

## 1. Answer B Concept 1

A teacher planning a six-week ecology unit notes that the classroom has four working microscopes for 28 students, the school is in a suburb with a large park nearby, ~~and the school mascot is a hawk~~. Which contextual-factors statement best meets the task?

**Target word:** a context statement with an implication **Decides:** A context fact must be tied to a planning decision.

|          |                   |                                                                                 |
|----------|-------------------|---------------------------------------------------------------------------------|
| A        | Camouflage        | Builds the plan on an irrelevant detail.                                        |
| <b>B</b> | <b>Correct</b>    | Each fact is paired with a specific instructional implication.                  |
| C        | Dropped element   | Lists facts without implications for instruction.                               |
| D        | Absolute language | Overreacts; the limit calls for structuring access, not removing hands-on work. |

## 2. Answer A Concept 2

A teacher ~~with ten years of experience~~ drafts goals for a unit on forces aligned to a middle school standard. ~~The unit will use toy cars borrowed from a colleague~~. Which goal is best written?

**Target word:** pick the measurable, aligned goal **Decides:** Observable performance combining a practice with a core idea.

|          |                   |                                                                                   |
|----------|-------------------|-----------------------------------------------------------------------------------|
| <b>A</b> | <b>Correct</b>    | Observable, joins a practice (investigation) with a core idea (force and motion). |
| B        | Confusion set     | "Understand" is not observable.                                                   |
| C        | Dropped element   | Recall only; leaves out the science practice named in the standard.               |
| D        | Adjacent category | An activity, not a learning outcome.                                              |

## 3. Answer D Concept 3

For a unit on chemical reactions ~~in a building renovated last summer~~, the pre-assessment asks students to balance equations and identify evidence of a chemical change. The draft post-assessment asks students to name lab safety equipment. ~~The unit runs four weeks~~. What is the main problem?

**Target word:** flaw in the assessment plan **Decides:** Pre and post measure different goals.

|          |                    |                                                                |
|----------|--------------------|----------------------------------------------------------------|
| A        | Adjacent category  | Length is not the issue; content alignment is.                 |
| B        | Inverted operation | The pre-assessment is what shows the starting point for gains. |
| C        | Absolute language  | The same scale does not make the measures comparable.          |
| <b>D</b> | <b>Correct</b>     | Alignment requires comparable measures of the same goals.      |

## 4. Answer D Concept 3 · transfer

A teacher's class scored much higher on the post-assessment than on the pre-assessment. Which fact would most weaken the claim that students learned the unit goals?

**Target word:** reverse: what undermines a gain claim **Decides:** Non-comparable assessments make gains uninterpretable.

|          |                    |                                                               |
|----------|--------------------|---------------------------------------------------------------|
| A        | Confusion set      | Timing is a minor factor compared with content comparability. |
| B        | Inverted operation | Identical or parallel items strengthen comparability.         |
| C        | Adjacent category  | Does not undermine the gain; it describes remaining need.     |
| <b>D</b> | <b>Correct</b>     | A gain could reflect easier items, not learning.              |

## 5. Answer B Concept 4

A teacher, ~~who shares a lab room with two colleagues,~~ drafts a density lesson: day 1, notes and vocabulary; day 2, worked examples; day 3, a lab where students float and sink objects to confirm the formula. Which revision best fits inquiry-based design?

**Target word:** sequence revision **Decides:** Explore before explain.

|          |                   |                                                              |
|----------|-------------------|--------------------------------------------------------------|
| A        | Absolute language | Explanation still has a place, after exploration.            |
| <b>B</b> | <b>Correct</b>    | Students explore the phenomenon before the teacher explains. |
| C        | Camouflage        | Solves the room detail, not the sequence problem.            |
| D        | Dropped element   | Leaves the confirm-at-the-end sequence unchanged.            |

## 6. Answer B Concept 4 • transfer

Students have just explored how the length of a pendulum affects its swing time and recorded patterns. In a 5E lesson, what should come next?

**Target word:** next phase after Explore **Decides:** Explain follows Explore.

|          |                    |                                         |
|----------|--------------------|-----------------------------------------|
| A        | Dropped element    | Skips Explain and Elaborate.            |
| <b>B</b> | <b>Correct</b>     | Explanation builds on exploration data. |
| C        | Confusion set      | Elaborate follows Explain.              |
| D        | Inverted operation | Engage comes before Explore.            |

## 7. Answer C Concept 5

During a unit on seasons, ~~taught in a room with a new interactive whiteboard,~~ many students write that summer happens because Earth is closer to the Sun. What is the most effective next step?

**Target word:** address the misconception **Decides:** Create evidence that contradicts the naive model.

|          |                   |                                                                       |
|----------|-------------------|-----------------------------------------------------------------------|
| A        | Dropped element   | Telling rarely changes a strongly held model.                         |
| B        | Camouflage        | Uses the tool but targets a different phenomenon.                     |
| <b>C</b> | <b>Correct</b>    | A discrepant event makes the misconception fail and prompts revision. |
| D        | Absolute language | Distance is not the cause of seasons.                                 |

## 8. Answer A Concept 6

Halfway through a unit on cells, a teacher ~~who teaches three sections a day~~ gives an exit ticket. 18 of 26 students confuse the cell wall with the cell membrane; the rest are secure. The pacing guide calls for starting genetics tomorrow. What is the best instructional decision?

**Target word:** respond to formative data **Decides:** Most students missed one specific goal.

|          |                    |                                                                |
|----------|--------------------|----------------------------------------------------------------|
| <b>A</b> | <b>Correct</b>     | Responds to the specific pattern and documents the adjustment. |
| B        | Inverted operation | Ignores clear evidence that most students missed the goal.     |
| C        | Absolute language  | Overcorrects; only one idea and 18 students need it.           |
| D        | Dropped element    | Measures again without changing instruction.                   |

### 9. Answer C Concept 6 · transfer

A teacher's unit report says, "On day 6, I added a second modeling activity on energy transfer." What must accompany this statement to show instructional decision-making?

**Target word:** what documents a decision **Decides:** A decision needs the evidence that triggered it and its effect.

|   |                   |                                                        |
|---|-------------------|--------------------------------------------------------|
| A | Adjacent category | Not evidence of a learning-based decision.             |
| B | Absolute language | An unsupported claim, not evidence.                    |
| C | <b>Correct</b>    | Decision-making is judged by evidence-to-action links. |
| D | Dropped element   | Shows what was planned, not why it changed.            |

### 10. Answer D Concept 7

A teacher ~~who runs the school garden club~~ analyzes a unit. Class pre-test mean: 35%; post-test mean: 78%. English learners (6 students): 30% to 60%. Other students (20): 37% to 83%. Which analysis is best supported?

**Target word:** subgroup interpretation **Decides:** English learners gained 30 points vs. 46 for peers.

|   |                   |                                                                |
|---|-------------------|----------------------------------------------------------------|
| A | Wrong stem number | That is the post-test mean, not the gain.                      |
| B | Absolute language | A class mean can hide subgroup gaps.                           |
| C | Dropped element   | Ignores starting points; compare gains, not only final scores. |
| D | <b>Correct</b>    | Compares gains across subgroups with the same measures.        |

### 11. Answer C Concept 7 · transfer

A class gained an average of 40 percentage points and scored 80% on the post-assessment. What was the pre-assessment average?

**Target word:** solve for the pre-test mean **Decides:** Pre = post – gain.

|   |                    |                                                        |
|---|--------------------|--------------------------------------------------------|
| A | Inverted operation | Adds the gain instead of subtracting.                  |
| B | Confusion set      | Treats the gain as a percent of the post-test (40/80). |
| C | <b>Correct</b>     | $80 - 40 = 40\%$ .                                     |
| D | Wrong stem number  | The post-test mean.                                    |

### 12. Answer A Concept 8

After teaching a unit ~~in a portable classroom with a noisy heater~~, a teacher drafts a reflection. The data show strong gains on the modeling goal but weak gains on the goal about designing controlled experiments. Which reflection best meets the task?

**Target word:** strongest reflection **Decides:** Links the weak goal to evidence and names a specific change and growth step.

|   |                   |                                                         |
|---|-------------------|---------------------------------------------------------|
| A | <b>Correct</b>    | Evidence, cause, concrete change, and a growth step.    |
| B | Dropped element   | Ignores the weak goal the data show.                    |
| C | Camouflage        | An irrelevant detail offered as the cause without data. |
| D | Adjacent category | Blames effort without evidence.                         |

## Triage grid

Write each item number in the box it belongs to.

|                       | <b>Right</b>                                          | <b>Wrong</b>                                                                          |
|-----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Sure</b>           | Owled.                                                | <b>Concept inversion — top priority.</b> A wrong mental model; it survives rewording. |
| <b>Unsure / guess</b> | Recognition, not mastery. Re-drill with the transfer. | Not learned yet.                                                                      |

## Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

| <b>Trap</b>        | <b>What it does</b>                                          | <b>In this drill</b> | <b>Your misses</b> |
|--------------------|--------------------------------------------------------------|----------------------|--------------------|
| Dropped element    | Leaves out a required term or step.                          | 9                    |                    |
| Wrong stem number  | A figure copied from the prompt that is not the answer.      | 2                    |                    |
| Adjacent category  | The right idea in the wrong bucket.                          | 5                    |                    |
| Inverted operation | Direction flipped: add/subtract, cause/effect, higher/lower. | 5                    |                    |
| Absolute language  | Always, never, guarantee — where judgment applies.           | 7                    |                    |
| Confusion set      | Every option is real; the test is exact role-matching.       | 4                    |                    |
| Camouflage         | Answers a stem fact that was never the question.             | 4                    |                    |

## Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

| # | Concept                                                      | Rule                                                                                                                                         | Trap                                                                                                         | Tell                                                                                  |
|---|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1 | <b>Contextual factors must shape the plan</b>                | Describe the class, community, and resources only to draw specific instructional implications from them.                                     | Listing demographics and facilities without saying how they change the unit.                                 | A context fact (limited equipment, reading levels, schedule) and a planning decision. |
| 2 | <b>Measurable, standards-aligned science goals</b>           | A learning goal names an observable student performance that joins a science practice to a core idea, aligned with a standard.               | Writing goals with unobservable verbs (understand, appreciate) or as activities.                             | Draft goals that use “understand,” “learn about,” or name a lab activity.             |
| 3 | <b>Pre- and post-assessment alignment</b>                    | The pre- and post-assessment must measure the same goals at comparable difficulty so a change in scores reflects learning.                   | Using a different or easier post-test, or a pre-test that measures something else.                           | A mismatch between what the pre-test and post-test ask.                               |
| 4 | <b>Inquiry sequencing (5E)</b>                               | Students engage and explore a phenomenon before the teacher explains; elaboration and evaluation follow.                                     | Front-loading lecture and vocabulary, then using the lab only to confirm.                                    | A lesson order that starts with notes and ends with a lab.                            |
| 5 | <b>Eliciting and addressing misconceptions</b>               | Probe prior ideas first, then create a situation that makes the misconception fail (a discrepant event) and let students revise their model. | Simply telling students the correct idea and moving on.                                                      | A common naive idea surfacing in student answers.                                     |
| 6 | <b>Instructional decision-making from formative evidence</b> | When mid-unit evidence shows many students missing a goal, adjust instruction for that goal before moving on, and document the reason.       | Staying on the pacing plan, or reteaching everything to everyone.                                            | An exit ticket or check showing a pattern of errors.                                  |
| 7 | <b>Analyzing student learning at three levels</b>            | Report whole-class gain, compare a meaningful subgroup, and examine individual students, using the same measures before and after.           | Reporting only the class post-test average, or drawing conclusions about a subgroup without comparing gains. | A data table with pre/post scores by group.                                           |
| 8 | <b>Evidence-based reflection</b>                             | Reflection links the most and least successful goals to specific evidence and names a concrete change and a professional growth step.        | Blaming student effort or writing general impressions without data.                                          | A reflection draft that says “students did not try hard enough” or “it went well.”    |

**Course scope (public description):** MA, Science Education Teacher Performance Assessment contains a comprehensive, original, research-based curriculum unit designed to meet an identified educational need. It provides direct evidence of the candidate’s ability to design and implement a multi-week, standards-based unit of instruction, assess student learning, and then reflect on the learning process. The WGU Teacher Performance Assessment requires students to plan and teach a multi-week standards-based instructional unit consisting of seven components: 1) contextual factors, 2) learning goals, 3) assessment, 4) design for instruction, 5) instructional decision-making, 6) analysis of student learning, and 7) self-evaluation and reflection.